

```
1 sns.FacetGrid(data, col='region', row='category', hue='status')
```

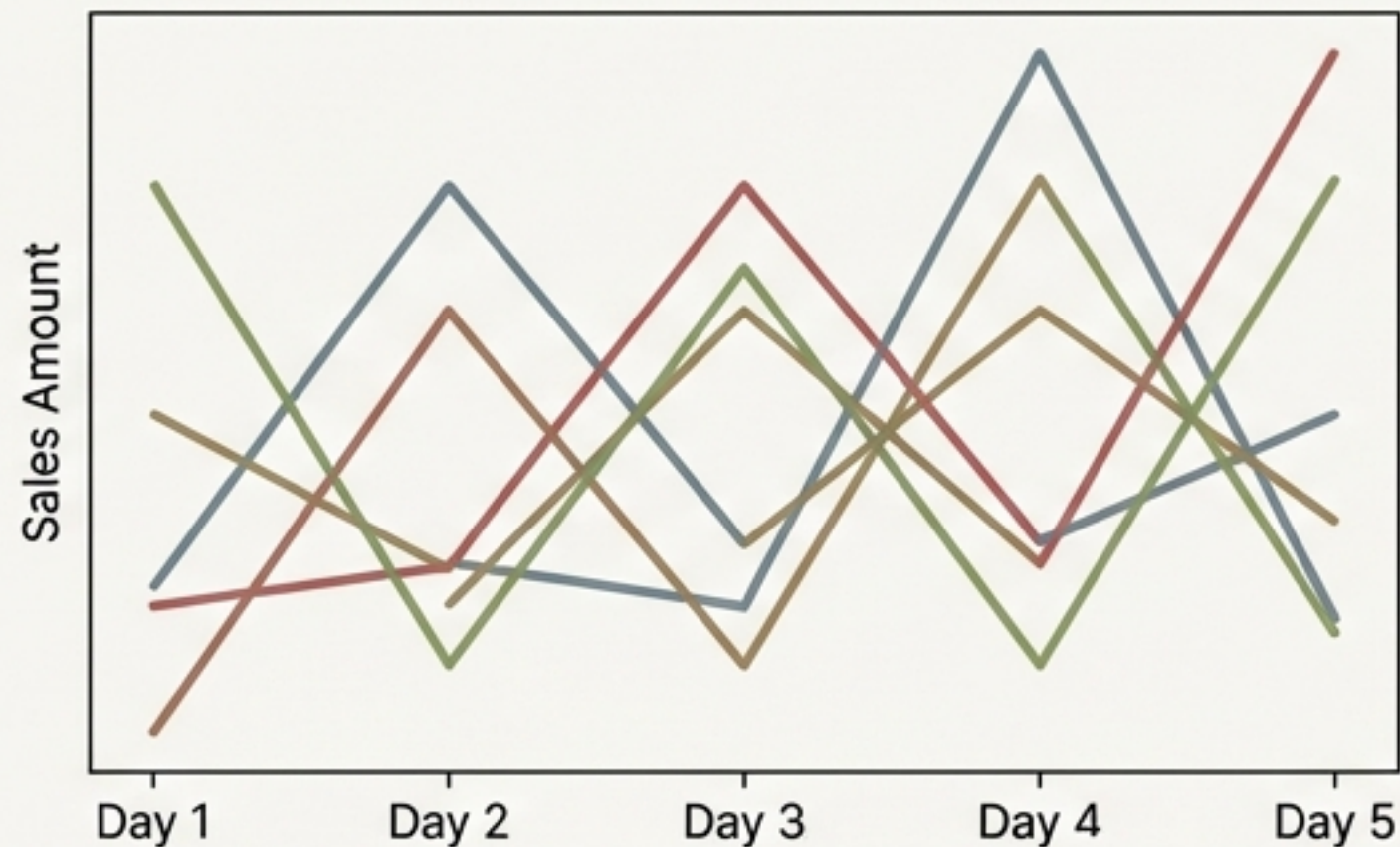
Mastering Facet Plots in Seaborn

The Art of Small Multiples for Clean Data Visualization

The Clutter Problem vs. The Small Multiples Solution

A Facet Plot (or **FacetGrid**) solves visual clutter by dividing a large dataset into small, individual sub-plots based on specific categories. Instead of forcing all data into one frame, it neatly organizes them side-by-side or top-to-bottom.

The Problem



Imagine plotting 5 days of sales for Laptops and Mobiles across Dhaka and Chittagong. 2 cities × 2 products = 4 overlapping lines in one messy chart.

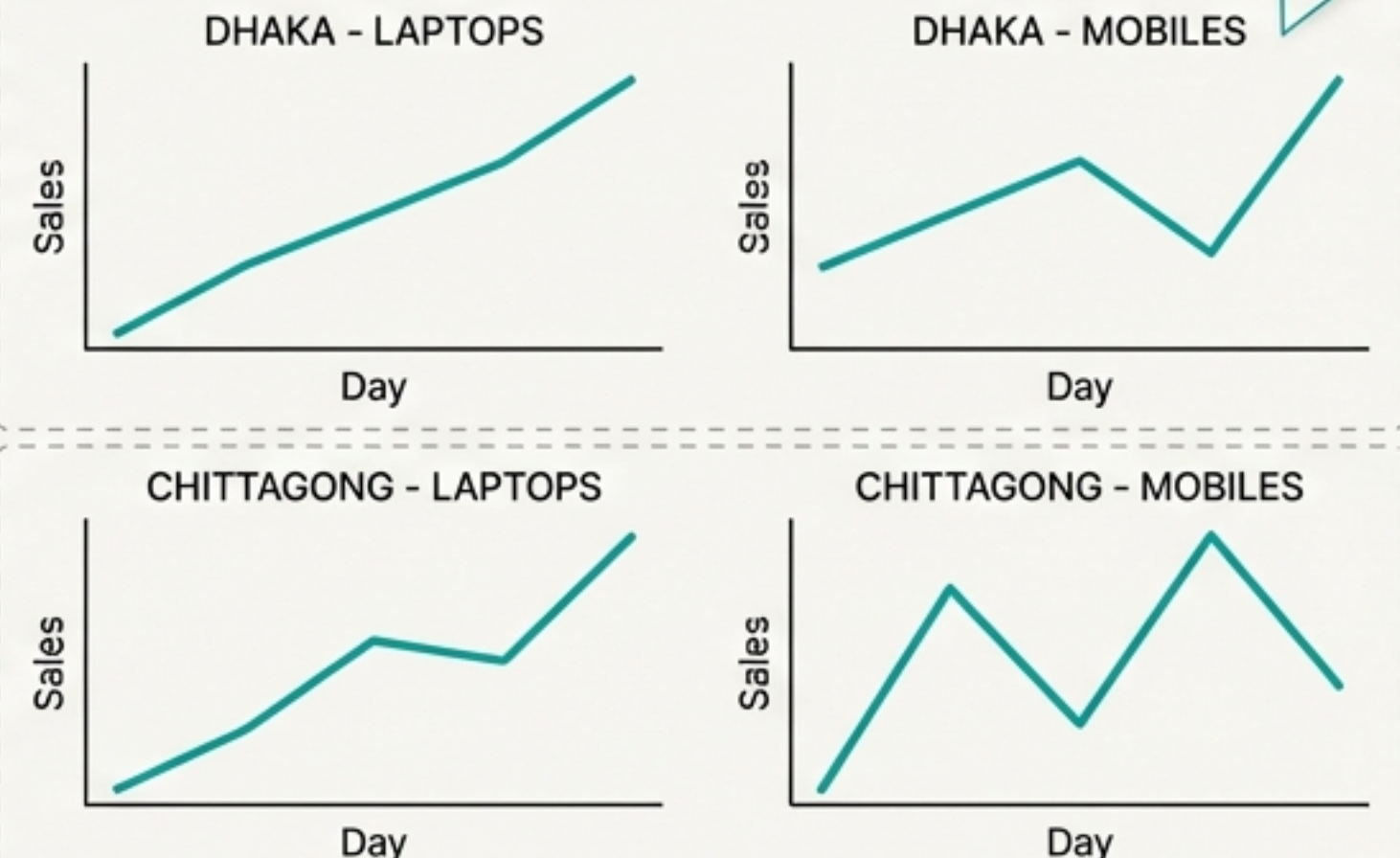
The Solution

```
facet_grid = (CITY, PRODUCT) => PLOT(SALES_OVER_TIME)
```

Row Facet: CITY

Column Facet: PRODUCT

Organized Structure:
4 clean, independent views.



Environment & Prerequisites

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

from google.colab import drive
drive.mount('/content/drive')
```

Data Manipulation
(pandas, numpy)

Visualization Engine
(seaborn, matplotlib, plotly)

File Storage
(Google Drive)

Standard data and visualization libraries loaded,
alongside Google Drive mounting for dataset access.

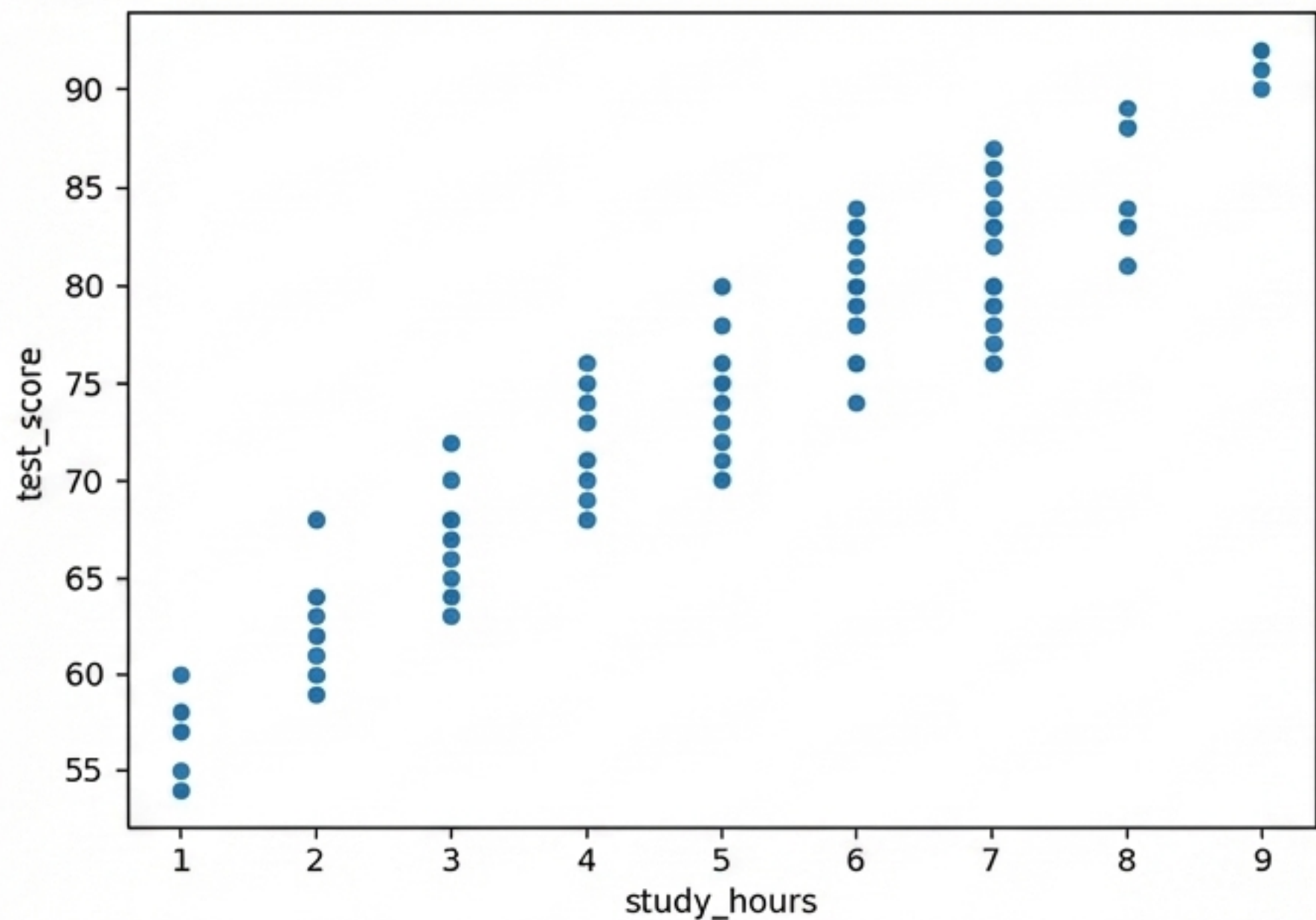
Loading the Dataset: sns_data.csv

```
student = pd.read_csv('/content/drive/MyDrive/Dataset/sns_data.csv')
```

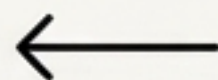
student_id	class_level	subject	study_hours	test_score	gender	hostel
1	Freshman	Math	5	78	Male	True
2	Freshman	Science	3	65	Female	False
...	...	...	...	...	...	...
92	Junior	Science	6	78	Female	False
95	Senior	Math	6	84	Male	False

The Baseline: A Standard Scatter Plot

```
sns.scatterplot(data=student, x="study_hours", y="test_score")
```



Insight Annotation



While this shows the overall positive trend between study hours and test scores, the granular categorical details—like differences between genders or living arrangements—are completely invisible.

The Facet Solution: Enter relplot

```
sns.relplot(  
    data=student,  
    x="study_hours",  
    y="test_score",  
    kind="scatter",  
    col="gender",  
    row="hostel"  
)
```

col (Column): Distributes the sub-plots horizontally (left to right) based on a category.

row (Row): Distributes the sub-plots vertically (top to bottom) based on a category.

Syntax Anatomy & Parameter Breakdown

data: Pulls directly from the loaded dataframe.

x & y: Maps the fundamental X and Y axes for every resulting sub-plot.

kind: Instructs Seaborn to render data points as dots rather than connecting lines.

```
sns.relplot(data=student,  
            x="study_hours",  
            y="test_score",  
            kind="scatter",  
            col="gender",  
            row="hostel")
```

col: The horizontal split. Creates a Left column (Male) and a Right column (Female).

row: The vertical split. Creates a Top row (hostel=False) and a Bottom row (hostel=True).

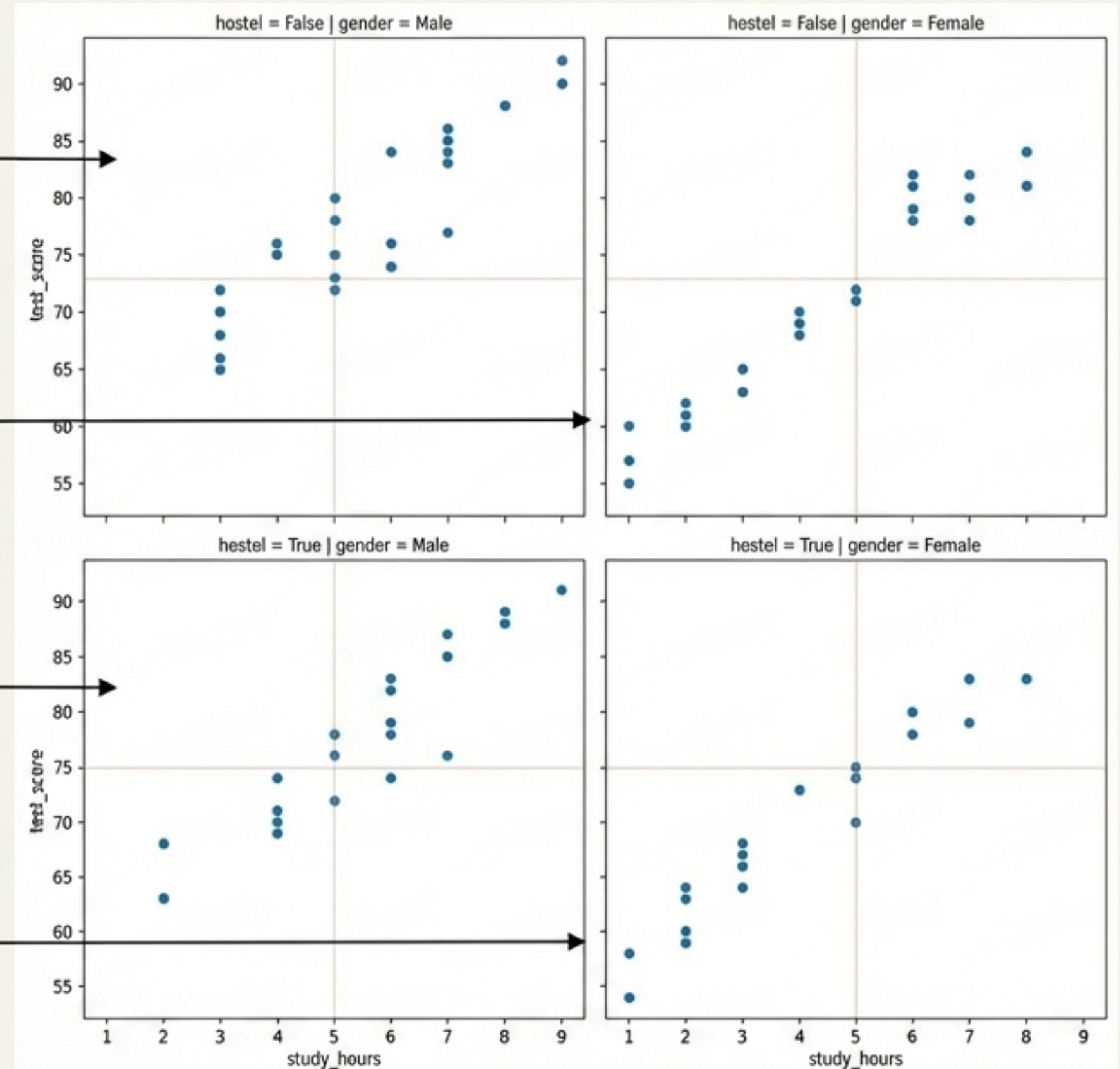
Deconstructing the 2x2 Grid Output

hostel = False | gender = Male:
Male students who do NOT live in a
hostel.

hostel = False | gender = Female:
Female students who do NOT live in a
hostel.

hostel = True | gender = Male:
Male students who DO live in a hostel.

hostel = True | gender = Female:
Female students who DO live in a hostel.



The Three Pillars of Faceting



Easy Comparison

Because every sub-plot shares the exact same X and Y axis scales, visual comparisons between groups are instant and mathematically reliable.



Finding Patterns

Isolating variables makes it incredibly easy to spot unique behaviors, trends, or outliers hidden within specific groups that aggregate graphs mask.



Clean View

It elegantly presents dense information without creating a cluttered, unreadable jumble of overlapping lines and overlapping data points.

Advanced Application: Wrapping & Color Coding

```
sns.relplot(data=student, x="study_hours", y="test_score",  
            kind="scatter", col="week", hue="gender", col_wrap=2)
```

Parameter Breakdown

- `hue="gender"`: Applies color-coding within the same plot instead of splitting.
- `col="week"`: Creates a column for every week.
- `col_wrap=2`: Prevents grid from becoming too wide.

